

```

*
*****
***

* PROGRAM: ACT_MainPaperTables_REPLICATION

* PURPOSE: This code replicates all main and appendix A tables and figures

*       for the published version of "Price Subsidies, Diagnostic Tests, and Targeting of
*       Malaria Treatment: Evidence from a Randomized Controlled Trial"

*
*****
****

```

* FILE NAMES

* MAIN DATASETS

```

global all_main ACT_AllMain_FINAL_pub
global all_ill_prob ACT_IllLvlMainWithMalProbs_FINAL_pub
global hh_follow_all ACT_HHFollowUp_All_FINAL_pub
global chem_main_long ACT_PharmLogPos_FINAL_pub
global baselines_mal ACT_BaselineMal_FINAL_pub
global pharma_txn ACT_NonProjectTxns_FINAL_pub

```

* put the file path to the replication data in quotation marks after the finaldata global

* put the file path to the FDO do file in quotation marks after the dofiles global

* note: you must end you file paths with a slash: / for Mac, \ for PC for hte FDR code to run

```

global finaldata "C:\Users\38164\Desktop\Cohen_AmEcoRev_2015_2lb5"

```

```

global dofiles "C:\Users\38164\Desktop\Cohen_AmEcoRev_2015_2lb5"

```

```
cd "$finaldata"
```

```
* SET ADDITIONAL CONTROLS FOR SPECS
```

```
global cont "B_head_age_imputed B_head_age_missing"
```

```
* YOU MUST SET THE doFDR GLOBAL TO 'Yes' AND RUN CODE CONTINUOUSLY FROM THE TOP ALL THE  
WAY DOWN TO
```

```
*      APPENDIX TABLE A7 TO CREATE APPENDIX TABLE A7! (Note -- in order to run tables one by one  
you
```

```
*      should set doFDR to 'No')
```

```
global doFDR "Yes"
```

```
* global doFDR "No"
```

```
clear
```

```
set more off
```

```
set mem 500m
```

```
set matsize 1000
```

```
mat fdr=J(1000,3,.)
```

```
    local ftable=2
```

```
    local frow=1
```

```
* *****
```

```
* TABLE 1 :: SUMMARY STATISTICS AND BALANCE CHECK
```

```
* *****
```

```
use $all_main, clear
```

```
set more off
```

```
mat t=J(43,7,.)
```

```
local row=1
```

```
foreach var in head_fem B_head_age head_edu head_lit head_mar subfarm head_dep B_hh_size  
B_adultteen head_acres B_dist_km ///
```

```
dist_clinic num_bednets share_undernet B_knowledge_correct B_heard_act B_act_best  
B_heard_rdt treat_h2o B_mal_ct B_antimal_cost {
```

```
local col=1
```

```
qui sum `var' if act500==1
```

```
mat t[`row',`col']= r(mean)
```

```
mat t[`row'+1,`col']= r(sd)
```

```
local ++col
```

```
cap reg `var' act40 act60 act100 rdt_any i.totstrata, r
```

```
foreach cov in act40 act60 act100 rdt_any {
```

```
mat t[`row',`col']= _b[`cov']
```

```
mat t[`row'+1,`col']= _se[`cov']
```

```
local ++col
```

```
}
```

```
test act40=act60=act100=rdt_any=0
```

```
mat t[`row',`col']= r(F)
```

```
mat t[`row'+1,`col']= r(p)
```

```
local ++col
```

```
mat t[`row',`col']= e(N)
```

```
local row=`row'+2
```

```
}
```

```
local col=1
```

```
foreach var in act500 act40 act60 act100 {
```

```

        replace `var`=0 if rdt_any==1
    }

foreach var in act500 act40 act60 act100 rdt_any {
    sum `var'
    mat t[`row',`col']= r(mean)*r(N)
    local ++col
}

mat li t

*
*****
*
* TABLE 2 :: IMPACT OF ACT AND RDT SUBSIDY ON CARE SEEKING, ACCESS
*
*****
*

use $all_ill_prob, clear

set more off

if "$doFDR" == "No" {
    local ftable=1
    local frow=1
}

keep if first_ep==1 & ex_post==0 & rdt_any==0

g byte ses_low= head_lit==0 if head_lit!=.
g young= LOG_patient_age<14
g act_any= act40 | act60 | act100

mat t= J(12,9,.)

local rowst=1

foreach group in all {

```

```
local col=1
```

```
    foreach out in took_act took_act_chem took_act_hc care_chem care_hc care_nothing  
took_maltest took_antibio {
```

```
    local row=`rowst'
```

```
* SPECIFICATION 1 (POOLED)
```

```
reg `out' act_any i.totstrata ex_post $cont if `group'==1, clu(househo)
```

```
    foreach v in act_any {
```

```
        mat t[`row',`col']=_b[`v']
```

```
        local ++row
```

```
        mat t[`row',`col']=_se[`v']
```

```
        local ++row
```

```
    if "`group'"=="all" {
```

```
        mat fdr[`frow',1]= `ftable'
```

```
        mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`v']/_se[`v']))
```

```
        mat fdr[`frow',3]=_b[`v']
```

```
        local ++frow
```

```
    }
```

```
}
```

```
* SPECIFICATION 2
```

```
reg `out' act40 act60 act100 i.totstrata ex_post $cont if `group'==1, clu(househo)
```

```
    foreach v in act40 act60 act100 {
```

```
        mat t[`row',`col']=_b[`v']
```

```
        local ++row
```

```
        mat t[`row',`col']=_se[`v']
```

```
        local ++row
```

```

if "`group'"=="all" {
    mat fdr[`frow',1]= `ftable'
    mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`v']/_se[`v']))
    mat fdr[`frow',3]= _b[`v']
    local ++frow
}
}

test act40=act60=act100=0
    mat t[`row',`col']= r(p)
    local ++row

test act40=act60=act100
    mat t[`row',`col']= r(p)
    local ++row

qui sum `out' if `group'==1 & act500 & e(sample)
    mat t[`row',`col']= r(mean)
    local ++row
    mat t[`row',`col']= e(N)

local ++col
}

local rowst=`row'+1
}

mat li t

*
*****
*

* TABLE 3 :: TARGETING RESULTS

```

```
*
*****
*
```

```
local ftable=3
```

```
if "$doFDR" == "No" local frow=1
```

```
use $all_ill_prob, clear
```

```
keep if ex_post==0 & rdt_any==0 & took_act==1 & act500==0
```

```
keep if first_ep==1
```

```
g group=2
```

```
tempfile endline
```

```
save "`endline'"
```

```
use $chem_main_long, clear
```

```
keep if ex_post==1 & used_act==1 & act500==0
```

```
drop if rdt_any==1
```

```
ren LOG_patient_age LOG_patient_age1
```

```
ren used_act_adult adult
```

```
g group=1
```

```
tempfile tomerge
```

```
save "`tomerge'"
```

```
use $all_main, clear
```

```
set more off
```

```
drop if rdt_any==1 | ex_post==0 | act500
```

```
keep if used_act==1
```

```
replace LOG_mal_prob21=. if rdt_pos==. | ex_post==0
```

```
ren used_act_adult adult
```

```
g group=0
```

```
append using ``tmerge''
```

```
append using ``endline''
```

```
replace all=1
```

```
g young= adult==0
```

```
g rdt_posA= rdt_pos if group==0
```

```
g rdt_posB= rdt_pos if group==1
```

```
g mal_probA= LOG_mal_prob21 if group==0
```

```
g mal_probB= LOG_mal_prob2 if group==1
```

```
g mal_probC= mal_prob2 if group==2
```

```
mat t=J(8,3,.)
```

```
local col=1
```

```
foreach take in rdt_posA mal_probA mal_probC {
```

```
    local row=1
```

```
    foreach type in all {
```

```
        xi: reg `take' act60 act100 if `type'==1, clu(househo)
```

```
        foreach var in act60 act100 {
```

```
            mat t[`row',`col']= _b[`var']
```

```
            local ++row
```

```
            mat t[`row',`col']= _se[`var']
```

```
            local ++row
```

```
            mat fdr[`frow',1]= `ftable'
```

```
            mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`var']/_se[`var'])))
```

```
            mat fdr[`frow',3]= _b[`var']
```

```
            local ++frow
```

```
        }
```

```

qui test act60=act100==0
mat t[`row',`col']= r(p)
    local ++row

qui test act60=act100
mat t[`row',`col']= r(p)
    local ++row

qui sum `take' if e(sample) & act40 & rdt_any==0
    mat t[`row',`col']= r(mean)
        local ++row
    mat t[`row',`col']= e(N)
        local ++row
}
local ++col
}

mat li t

* *****
* TABLE 4 :: MECHANISMS
* *****

local ftable=4
    if "$doFDR" == "No" local frow=1
use $all_main, clear
set more off
drop if act500 | rdt_any==1
    g used_act_young= used_act_baby+used_act_kid+used_act_teen
    g frac_adult= used_act_adult if used_act==1
    g rdt_pos_young= rdt_pos if ex_post & used_act_young==1

```

```

g rdt_pos_adult= rdt_pos if ex_post & used_act_adult==1
g finalout= rdt_pos if ex_post & used_act==1

local col=1
mat t=J(7,4,.)
foreach out in used_act_young used_act_adult rdt_pos_young rdt_pos_adult {
local cme "i.totstrata $cont"

if "`out'"=="rdt_pos_adult" | "`out'"=="rdt_pos_young" local cme ""
local row=1
reg `out' act60 act100 `cme', r

foreach price in act60 act100 {
mat t[`row',`col']= _b[`price']
local ++row
mat t[`row',`col']= _se[`price']
local ++row

mat fdr[`frow',1]= `ftable'
mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`price']/_se[`price']))
mat fdr[`frow',3]= _b[`price']
local ++frow
}

test act60=act100=0
mat t[`row',`col']= r(p)
local ++row
qui sum `out' if e(sample) & act40
mat t[`row',`col']= r(mean)
local ++row
mat t[`row',`col']= e(N)
local ++row

```

```

        local ++col
    }
mat li t

*
*****
*

* TABLE 5 :: IMPACT OF RDT SUBSIDY ON CARE SEEKING, ACCESS

*
*****
*

local ftable=5

    if "$doFDR" == "No" local frow=1
use $all_ill_prob, clear
    set more off
    keep if first_ep==1
    drop if act500
    keep if rdt_any==1 | ex_post==0

foreach p in act40 act60 act100 {
    g R`p`= rdt_any*`p'
}

mat t= J(17,9,.)
local rowst=1
foreach group in all {
    local col=1
        foreach out in care_chem care_hc care_nothing took_maltest took_rdt took_micro took_act
took_antibio {
            local row=`rowst'

```

* SPECIFICATION 1

```
reg `out' rdt_any i.totstrata $cont if `group'==1, clu(househo)

foreach v in rdt_any {

    mat t[`row',`col']= _b[`v']

    local ++row

    mat t[`row',`col']= _se[`v']

    local ++row


    if "`group'"=="all" {

        mat fdr[`frow',1]= `ftable'

        mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`v']/_se[`v']))

        mat fdr[`frow',3]= _b[`v']

        local ++frow

    }

}

qui sum `out' if rdt_any==0 & e(sample)

mat t[`row',`col']= r(mean)

local ++row

mat t[`row',`col']= e(N)

local ++row
```

* SPECIFICATION 2

```
reg `out' Ract40 Ract60 Ract100 act60 act100 i.totstrata $cont if `group'==1, clu(househo)

foreach v in Ract40 Ract60 Ract100 act60 act100 {

    mat t[`row',`col']= _b[`v']

    local ++row

    mat t[`row',`col']= _se[`v']

    local ++row
```

```

if "`group'"=="all" {
    mat fdr[`frow',1]= `ftable'
    mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`v']/_se[`v']))
    mat fdr[`frow',3]= _b[`v']
    local ++frow
}

}

test Ract40=Ract60=Ract100=0
mat t[`row',`col']= r(p)
local ++row

qui sum `out' if `group'==1 & rdt_any==0 & e(sample) & act40
mat t[`row',`col']= r(mean)
local ++row
mat t[`row',`col']= e(N)

local ++col
}

local rowst=`row'+1
}

mat li t

```

```

* ****

```

```

* TABLE 6: TARGETING TABLE FOR RDTS

```

```

* ****

```

```

local ftable=6

```

```

if "$doFDR" == "No" local frow=1

```

```

use $all_main, clear

```

```

drop if act500

keep if ex_post==1

g rdt_pos_trt= rdt_pos if sought_treat==1
g used_rdt_trt= used_rdt if sought_treat==1
g comp_neg= used_act==0 if used_rdt==1 & rdt_pos==0
g comp_pos= used_act==1 if used_rdt==1 & rdt_pos==1
g rdt_pos_act= rdt_pos if used_act==1

foreach var in used_rdt_trt comp_neg comp_pos {
    replace `var'=. if rdt_any==0
}

foreach p in act40 act60 act100 {
    g R`p'= rdt_any*`p'
}

local rowct=1
local col=1
mat t=J(10,4,.)
foreach group in all {
    foreach out in sought_treat rdt_pos_trt rdt_pos_act {
        ren `out' `out'_all
        local row=`rowct'

        local e1 ""
        if ""`out'""=="sought_treat" local e1 "i.totstrata $cont"

        reg `out'_`group' rdt_any act60 act100 `e1', r
        mat t[`row',`col']=_b[rdt_any]
    }
    local rowct+=1
    local col+=1
}

```

```

        local ++row
mat t[`row',`col']= _se[rdt_any]

        local ++row
if "`group'"=="all" {
        mat fdr[`frow',1]= `ftable'
        mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[rdt_any]/_se[rdt_any]))
        mat fdr[`frow',3]= _b[rdt_any]
        local ++frow
}

reg `out'_`group' Ract40 Ract60 Ract100 act60 act100 `e1', r

foreach cov in Ract40 Ract60 Ract100 {
        mat t[`row',`col']= _b[`cov']
        local ++row
        mat t[`row',`col']= _se[`cov']
        local ++row
if "`group'"=="all" {
        mat fdr[`frow',1]= `ftable'
        mat fdr[`frow',2]= 2*ttail(e(df_r),abs(_b[`cov']/_se[`cov']))
        mat fdr[`frow',3]= _b[`cov']
        local ++frow
}
}

qui sum `out'_`group' if e(sample) & rdt_any==0 & act40
        mat t[`row',`col']= r(mean)
        local ++row
        mat t[`row',`col']= e(N)
        local ++row

ren `out'_all `out'

local ++col

```

```

}

foreach out in used_rdt_trt {
    ren `out' `out'_all
    local row=`rowct'

    sum `out'`_group' if rdt_any==1
        mat t[`row',`col']= r(mean)
        local ++row
        local ++row

    foreach val in act40 act60 act100 {
        sum `out'`_group' if rdt_any==1 & `val'==1
            mat t[`row',`col']= r(mean)
            local ++row
            local ++row
        }
        local ++row
        qui sum `out'`_group' if rdt_any==1
        mat t[`row',`col']= r(N)
    ren `out'_all `out'
    local ++col
}
local rowct=`row'+1
local col=1
}

mat li t

* ****

```

* TABLE 7 :: T/OT/UT ESTIMATES -- EXTRA ESTIMATES

* note: the calculations for this table are done in excel. information in other

* tables can be combined with this information to create the estimates in

* Table 7

* ****

set more off

mat t=J(12,3,.)

use \$all_ill_prob, clear

set more off

keep if first_ep==1

keep if ex_post==0 | rdt_any==1

replace ex_post=0

foreach p in act40 act60 act100 {

g R`p`= rdt_any*`p`

}

local col=1

foreach out in took_act_hc took_act_chem {

local row=1

reg `out' Ract40 Ract60 Ract100 act60 act100 i.totstrata \$cont if act500==0, clu(househo)

foreach v in Ract40 Ract60 Ract100 act60 act100 {

mat t[`row',`col']=_b[`v']

local ++row

mat t[`row',`col']=_se[`v']

local ++row

}

qui sum `out' if act40 & rdt_any==0 & e(sample)

mat t[`row',`col']= r(mean)

local ++row

mat t[`row',`col']= e(N)

```

                                local ++row
                                local ++row
                                local ++col
                                }

use $all_main, clear

                                keep if used_act==1 & (act40 | act60 | act100) & checklistonly==0 & ex_post==1
                                foreach price in act40 act60 act100 {
                                        g `price'_rdt= `price'*rdt_any
                                }

                                local row=1
                                reg rdt_pos act60 act100 act40_rdt act60_rdt act100_rdt, r
                                foreach cov in act60 act100 act40_rdt act60_rdt act100_rdt {
                                        mat t[`row',`col']= _b[`cov']
                                        local ++row
                                        mat t[`row',`col']= _se[`cov']
                                        local ++row
                                }
                                mat t[`row',`col']= _b[_cons]
                                local ++row
                                mat t[`row',`col']= e(N)

mat li t

```

```

* *****

```

```

* TABLE A1 :: BASELINE TREATMENT SEEKING BEHAVIOR

```

```

* *****

```

```

use $all_main, clear

```

```

set more off

```

```

g mal_kids= B_mal_baby_hh+B_mal_kid_hh+B_mal_teen
g mal_adults= B_mal_adult
g byte ep_kids= mal_kids>0 if mal_kids!=.
g byte ep_adults= mal_adults>0 if mal_adults!=.

```

```

mat t=J(18,4,.)

* FIRST TWO ROWS: AT HOUSEHOLD LEVEL

replace base_rdt=. if B_mal_episode==0
replace base_micro=. if B_mal_episode==0

local col=1

foreach group in all {
    local row=1
    foreach var in B_mal_ct B_mal_episode base_rdt base_micro {
        sum `var' if `group'
        mat t[`row',`col']= r(mean)
        local ++row
    }
    local ++col
}

foreach var in kids adults {
    local row=1
    foreach st in mal ep {
        sum `st'_`var'
        mat t[`row',`col']= r(mean)
        local ++row
    }
    local ++col
}

keep all househo

```

```

    sort househo
    tempfile tomerge
    save "`tomerge'"
use $baselines_mal, clear
    drop if malaria==0
        sort househo
        merge househo using "`tomerge'"
        tab _merge
        drop if _merge==2 /* not in episode ledger */
        drop _merge
local row=6
    g test= adult+teen+kid+baby
    foreach var in adult teen kid baby {
        replace `var'= . if test==0 & b5_==. & b6_==.
    }
replace took_other=0 if forgot_name==1
foreach var in took_act took_sp took_aq took_other forgot_name {
    replace `var'=0 if `var'==. & no_antimal==1
}

g byte care_none= diag_init==4 if diag_init!=.
g byte care_hc= diag_init==1 | diag_init==2 if diag_init!=.
g byte care_chem= diag_init==3 if diag_init!=.

    replace care_hc=1 if care_none==1 & (got_drugs==1 | got_drugs==2)
    replace care_chem=1 if care_none==1 & got_drugs==3
    replace care_hc=. if care_none==1 & got_drugs==5
    replace care_chem=. if care_none==1 & got_drugs==5
    replace care_none=. if care_none==1 & got_drugs==5
    replace care_none=0 if got_drugs==1 | got_drugs==2 | got_drugs==3 | got_drugs==4

```

```

replace care_hc=1 if care_none==0 & care_hc==. & got_drugs==1
replace care_chem=1 if care_none==0 & care_chem==. & got_drugs==3
replace antimal_cost=. if got_drugs==.

foreach var in drugs_public drugs_chemist drugs_oth {
    replace `var'= . if no_antimal==1
}

foreach var in care_none care_hc care_chem no_antimal took_act took_sp took_aq took_oth
forgot_name drugs_public drugs_chemist drugs_oth antimal_cost {
    sum `var'
    mat t[`row',1]= r(mean)
    local col=2

foreach sub in adult {
    reg `var' `sub', clu(househo)
    mat t[`row',`col']= _b[_cons]
    local ++col
    mat t[`row',`col']=_b[_cons]+_b[`sub']
    local ++col
    test `sub'
    mat t[`row',`col']= r(p)
    local ++col
}
local ++row
}

mat li t

```

*

*

* TABLE A2 :: REPORTING BIAS WITH ENDLINE ILLNESS EPISODES

```

*
*****
*

```

```

use $all_main, clear

    set more off

    g ill_hh= E_tabJ_illness>0 if E_tabJ_illness!=.

    g XE_tabJ_illness=E_tabJ_illness

    keep househo ill_hh XE_tabJ_illness act40 act60 act100 rdt_any ex_post totstrata $cont
headage_x head_lit

    tempfile appender

    save "`appender'"

```

```

use $all_ill_prob, clear

*set more off

append using "`appender'"

    g byte act40100= coartemprice<500

    g f_LOG_patient_age= LOG_patient_age if first_ep==1

    g mal_prob_hh= mal_prob2 if first_ep==1

    replace max_length=. if first_ep==0

    g byte ses_low = head_lit==0

    replace all=1

mat t=J(13,5,.)

local rowct=1

foreach group in all {

    local col=1

    foreach out in ill_hh XE_tabJ_illness mal_prob_hh max_length f_LOG_patient_age {

        if "`out'"!="ill_hh" & "`out'"!="XE_tabJ_illness" replace `out'=. if mal_prob2==.

        local row= `rowct'

        xi: reg `out' act40 act60 act100 rdt_any ex_post i.totstrata $cont if `group'==1, clu(househo)

        foreach var in act40 act60 act100 rdt_any ex_post {

```

```

        mat t[`row',`col']= _b[`var']

        local ++row

        mat t[`row',`col']= _se[`var']

        local ++row

    }

    test act40=act60=act100

    mat t[`row',`col']= r(p)

    local ++row

qui sum `out' if e(sample)

    mat t[`row',`col']= r(mean)

    local ++row

    mat t[`row',`col']= e(N)

    local ++row

    local ++col

}

local rowct= `row'

}

mat li t

```

```

* *****

```

```

* TABLE A3 :: PREDICTING MALARIA POSITIVITY -- PROBIT MFX

```

```

* *****

```

```

use $hh_follow_all, clear

```

```

    keep if days_ago<=3

```

```

probit rdt_pos cough chills headache diarrhea runnynose vomit bodypain malaise appetite old2
LOG_patient_age age_sq cough_old2-appetite_old2 LOG_patient_age_old2 age_sq_old2, vce(cluster
househo)

```

mfx compute

```
* *****  
  
* TABLE A4 :: RDT TAKEUP BY RDT PRICE  
  
* *****  
  
use $all_main, clear  
  
set more off  
  
    drop if coartemprice>100  
  
  
g used_rdt_c= used_rdt if sought_treat==1  
mat t=J(11,3,.)  
local col=1  
foreach out in used_rdt sought_treat used_rdt_c {  
    local row=1  
    xi: reg `out' rdt_free rdt_15r rdt_15 act60 act100 ex_post i.totstrata, r  
        foreach coeff in rdt_free rdt_15r rdt_15 {  
            mat t[`row',`col']= _b[`coeff']  
            local ++row  
            mat t[`row',`col']= _se[`coeff']  
            local ++row  
        }  
    test rdt_free=rdt_15r=rdt_15  
    mat t[`row',`col']= r(p)  
    local ++row  
    xi: reg `out' rdt_any act60 act100 ex_post i.totstrata , r  
        mat t[`row',`col']= _b[rdt_any]  
        local ++row  
        mat t[`row',`col']= _se[rdt_any]
```

```

                                local ++row
qui sum `out' if e(sample) & rdt_none==1
                                mat t[`row',`col']= r(mean)
                                local ++row
                                mat t[`row',`col']= e(N)
                                local ++col
                                }
mat li t

* *****
* TABLE A5: TARGETING WITH RDTs
* *****

use $all_ill_prob, clear
set more off
drop if (ex_post & rdt_any==0) | act500
keep if took_antimal==1 | took_mal==1
                                ren LOG_patient_age LOG_patient_age1

ren mal_prob2 LOG_mal_prob21
replace sought_treat= took_antimal | took_mal
replace used_act= took_antimal
replace used_rdt= took_mal

keep if first_ep | second_ep
g dlong=3
tempfile tomerge3
                                save "`tomerge3'"

```

```

    keep if first_ep
        replace dlong=2
tempfile tomerge2
    save "`tomerge2'"

use $chem_main_long, clear
    keep if used_act==1 | used_rdt==1
    drop if act500

    ren LOG_patient_age LOG_patient_age1
    ren LOG_mal_prob2 LOG_mal_prob21
    g dlong=1
tempfile tomerge
    save "`tomerge'"

use $all_main, clear
    drop if act500 | checklistonly
    g dlong=0

append using "`tomerge2'"
append using "`tomerge3'"
append using "`tomerge'"

replace rdt_pos= LOG_mal_prob21
g rdt_pos_trt= rdt_pos if sought_treat==1
g rdt_pos_act= rdt_pos if used_act==1

foreach p in act40 act60 act100 {
    g R`p`= rdt_any*`p'

```

```
}
```

```
local rowct=1
```

```
local col=1
```

```
mat t=J(10,4,.)
```

```
foreach group in 0 2 {
```

```
  foreach out in rdt_pos_trt rdt_pos_act {
```

```
    ren `out' `out'_all
```

```
    local row=`rowct'
```

```
    reg `out' rdt_any act60 act100 if dlong==`group', clu(household)
```

```
      mat t[`row',`col']=_b[rdt_any]
```

```
      local ++row
```

```
      mat t[`row',`col']=_se[rdt_any]
```

```
      local ++row
```

```
    reg `out' Ract40 Ract60 Ract100 act60 act100 if dlong==`group', clu(household)
```

```
      foreach cov in Ract40 Ract60 Ract100 {
```

```
        mat t[`row',`col']=_b[`cov']
```

```
        local ++row
```

```
        mat t[`row',`col']=_se[`cov']
```

```
        local ++row
```

```
      }
```

```
    qui sum `out' if e(sample) & rdt_any==0 & act40
```

```
      mat t[`row',`col']=r(mean)
```

```
      local ++row
```

```
      mat t[`row',`col']=e(N)
```

```
      local ++row
```

```
  ren `out'_all `out'
```

```
local ++col  
}  
}
```

mat li t

* ****

* APPENDIX TABLE A7

* note, you MUST

* A) run the code WITHOUT BREAKS from start to finish and

* B) set the global doFDR to "Yes"

* in order to successfully execute this code

* ****

drop_all

```
svmat fdr
```

```
ren fdr2 pval
```

```
ren fdr1 table
```

```
drop if pval==.
```

```
do "{dofiles}fdr_sharpened_qvalues.do"
```

```
keep if pval<=.1
```

```
ren pval orig_pval
```

```
ren fdr3 coeff
```

```
ren bky q_value
```

```
list table coeff orig_pval q_value, noobs clean
```

* ****

* ****

* FIGURES

* ****

* ****

* ****

* FIGURE 3 :: POSITIVITY BY AGE

* ****

use \$all_main, clear

keep if (act40 | act100 | act60) & ex_post==1 & rdt_any==0 & used_act==1 & checklistonly==0

keep rdt_pos LOG_patient_age1

g samp="CHEMIST"

tempfile add2

save "`add2'"

use \$hh_follow_all, clear

g samp="HH"

* ONLY OBS W/ MAL TEST RESULTS

keep if rdt_pos!=.

* LIMIT SAMPLE TO EPS THAT STARTED THREE DAYS AGO OR LESS

keep if days_ago<=3

append using "`add2'"

foreach var in LOG_patient_age LOG_patient_age1 {

replace `var'=60 if `var'>60 & `var'!=.

}

lpoly rdt_pos LOG_patient_age, deg(1) nscatter ci tit("A. All Illnesses", size(medium))
ylab(0(.2).8) xline(14, lcol(gs10)) ///

legend(off) graphregion(color(white) fcolor(white)) name(PA, replace) ytit("Share
Positive") xtit("Age") ///

```

xlab(0 14 20 40 60) note(" ") generate(pred_general) at(LOG_patient_age)
se(se_general)

lpoly rdt_pos LOG_patient_age1 if samp=="CHEMIST", deg(1) ci nscatter tit("B. Retail-Sector
Subsidized ACT Takers", size(medium)) xlab(0 14 20 40 60) ///

legend(off) xline(14, lcol(gs10)) graphregion(color(white) fcolor(white)) ylab(0(.2).8)
ytit("Share Positive") ///

note(" ") xtit("Age") name(PB, replace) generate(pred_chem) at(LOG_patient_age)
se(se_chem)

```

```

foreach var in general chem {
    g ul_`var'= se_`var'*1.96 + pred_`var'
    g ll_`var'= pred_`var'- se_`var'*1.96
}

gr combine PA PB, xsize(10) ysize(6) graphregion(color(white) fcolor(white)) ycommon

```

* ****

* FIGURE 4: DEMAND CURVES (ALL)

* APPENDIX FIGURE A4: DEMAND CURVES BY AGE

* ****

```

foreach age in all young adult {
    local tit "title(" ")
    if "`age'"=="young" local tit "title("I. Under Age 14", size(medsmall))"
    if "`age'"=="adult" local tit "title("II. Ages 14 and Older", size(medsmall))"

    local ylab2 "ylab(0(.1).5)"
    local ylab "ylab(0(.2).8)"

    if "`age'"=="young" | "`age'"=="adult" {
        local ylab "ylab(0(.1).5)"
        local ylab2 "ylab(0(.1).6)"
    }
}

```

```

    }
use $all_main, clear
    set more off
    drop if checklsto
    drop if rdt_any==1 | ex_post==1
qui tab totstrata, ge(strdum)
    foreach var of varlist $cont strdum* {
        qui sum `var'
        replace `var' = `var' - r(mean)
    }
replace num_actv=0 if num_actv==.

g used_act_young= used_act_baby + used_act_kid + used_act_teen + ///
    used_act_baby2 + used_act_kid2 + used_act_teen2 + ///
    used_act_baby3 + used_act_kid3 + used_act_teen3 + ///
    used_act_baby4 + used_act_kid4 + used_act_teen4
replace used_act_adult= used_act_adult + used_act_adult2 + used_act_adult3 + used_act_adult4

g used_act_all = used_act + used_act2 + used_act3 + used_act4

mat t=J(10,16,.)
    local row=1
    reg used_act `age' act40 act60 act100 act500 ex_post $cont strdum1-strdum28, nocons r
    test act40=act100
    test act40=act60
    test act60=act100
    test act40=act60=act100
    foreach price in 40 60 100 500 {
        mat t[`row',1]= `price'
    }

```

```

mat t[`row',2]= _b[act`price']
mat t[`row',3]= _b[act`price'] - 1.96*_se[act`price']
mat t[`row',4]= _b[act`price'] + 1.96*_se[act`price']
local ++row
}

```

use \$all_ill_prob, clear

```

drop if ex_post==1 | illness==0
drop if rdt_any==1
keep if first_ep==1
replace used_act_v=0 if num_actv==0 | num_actv==.
g young= LOG_patient_age<14

```

qui tab totstrata, ge(strdum)

```

foreach var of varlist $cont strdum* {
    qui sum `var'
    replace `var'= `var'-r(mean)
}

```

local row=1

reg took_act act40 act60 act100 act500 ex_post \$cont strdum1-strdum28 if `age'==1, nocons
clu(househo)

```

foreach price in 40 60 100 500 {
    mat t[`row',5]= `price'
    mat t[`row',6]= _b[act`price']
    mat t[`row',7]= _b[act`price'] - 1.96*_se[act`price']
    mat t[`row',8]= _b[act`price'] + 1.96*_se[act`price']
    local ++row
}

```

```

local row=1

reg used_act_v act40 act60 act100 act500 ex_post $cont strdum1-strdum28 if `age'==1, nocons
clu(househo)

foreach price in 40 60 100 500 {
    mat t[`row',9]= `price'
    mat t[`row',10]= _b[act`price']
    mat t[`row',11]= _b[act`price'] - 1.96*_se[act`price']
    mat t[`row',12]= _b[act`price'] + 1.96*_se[act`price']
    local ++row
}

use $all_ill_prob, clear

drop if rdt_any==1 | ex_post==1

replace used_act_v=0 if num_actv==0 | num_actv==.

g young= LOG_patient_age<14
g used_act_v_young= used_act_v*young
replace used_act_v_adult= used_act_v*adult
g totadmin= used_act + used_act2 + used_act3 + used_act4

collapse (sum) used_act_v used_act_v_young used_act_v_adult (mean) num_actv act40 act60
act100 act500 totadmin $cont totstrata ex_post, by(househo)

g anyuse_all= used_act_v if used_act_v!=.
g anyuse_young= used_act_v_young if used_act_v!=.
g anyuse_adult= used_act_v_adult if used_act_v!=.

qui tab totstrata, ge(strdum)

foreach var of varlist $cont strdum* {
    qui sum `var'

```

```

        replace `var' = `var' - r(mean)
    }

    local row=1

    reg anyuse_`age' act40 act60 act100 act500 ex_post $cont strdum1-strdum28, nocons
    clu(househo)

    test act40=act100

    test act40=act60

    test act60=act100

    test act40=act60=act100

    foreach price in 40 60 100 500 {

        mat t[`row',13]= `price'

        mat t[`row',14]= _b[act`price']

        mat t[`row',15]= _b[act`price'] - 1.96*_se[act`price']

        mat t[`row',16]= _b[act`price'] + 1.96*_se[act`price']

        local ++row

    }

drop _all

svmat t

* ENDLINE

twoway (scatter t10 t6 t1, connect(direct direct) lcol(gs7 black) mcol(gs7 black)) (rcap t11 t12 t1,
lcol(gs7)) (rcap t7 t8 t1, lcol(black)), ///

        xtit("Subsidy Level", size(small)) xlab(40 "92%" 60 "88%" 100 "80%" 500 "None",
labsize(vsmall)) name(PA, replace) tit("A. ACT Treatment for First Endline Illness Episodes",
size(medsmall)) ///

        graphregion(color(white) fcolor(white)) `ylab2' legend(order(1 2) size(small) label(1
"Used ACT Voucher") label(2 "Took Any ACT"))

* CHEM LOG

```

```

twoway      (scatter t2 t14 t1, connect(direct direct) lcol(gs7 black) mcol(gs7 black)) (rcap t3
t4 t1, lcol(gs7)) (rcap t15 t16 t1, lcol(black)), ///

      xtfit("Subsidy Level", size(small)) xlab(40 "92%" 60 "88%" 100 "80%" 500 "None",
labsize(vsmall)) name(PB, replace) tit("B. Number ACT Vouchers Redeemed", size(medsmall)) ///

      graphregion(color(white) fcolor(white)) `ylab' legend(order(1 2) size(small) label(1
"Admin. Data") label(2 "Endline Data"))

gr combine PA PB, row(2) col(1) graphregion(color(white) fcolor(white)) xsize(7.5) ysize(10) name(`age',
replace) `tit'

}

```

```

* ****

```

```

* FIGURE 5: RDT COMPLIANCE AND THE SUBSIDY LEVEL

```

```

* ****

```

```

use $all_main, clear

```

```

    set more off

```

```

    drop if act500

```

```

    drop if checklistonly

```

```

    keep if rdt_any==1

```

```

    g no_act= used_act==0

```

```

mat t=J(3,6,.)

```

```

local row=1

```

```

local col=1

```

```

* COMPLY WITH NEGATIVE

```

```

reg no_act act40 act60 act100 if used_rdt==1 & rdt_pos==0, r nocons

```

```

    foreach out in act40 act60 act100 {

```

```

        mat t[`row',`col']= _b[`out']

```

```

        mat t[`row',`col'+1]= _b[`out']-1.96*_se[`out']

```

```

        mat t[`row',`col'+2]= _b[`out']+1.96*_se[`out']

```

```

        local ++row
    }

```

```

    }

    local col=`col'+3

    local row=1

* COMPLY WITH POSITIVE

reg used_act act40 act60 act100 if used_rdt==1 & rdt_pos==1, r nocons

    foreach out in act40 act60 act100 {

        mat t[`row',`col']=_b[`out']

        mat t[`row',`col'+1]=_b[`out']-1.96*_se[`out']

        mat t[`row',`col'+2]=_b[`out']+1.96*_se[`out']

        local ++row

    }

drop _all

svmat t

g priceg=_n

twoway (bar t1 priceg, barw(.8)) (rcap t2 t3 priceg), ylab(0(.2)1) xlab(1 "92%" 2 "88%" 3 "80%")
ytit("Compliance Rate") name(PA, replace) ///

    tit("A. Complied - Negative Test (Did Not Take ACT)", size(medsmall)) legend(off) xtit("Subsidy
Level") graphregion(color(white) fcolor(white))

twoway (bar t4 priceg, barw(.8)) (rcap t5 t6 priceg), ylab(0(.2)1) xlab(1 "92%" 2 "88%" 3 "80%")
ytit("Compliance Rate") name(PB, replace) ///

    tit("B. Complied - Positive Test (Took ACT)", size(medsmall)) legend(off) xtit("Subsidy Level")
graphregion(color(white) fcolor(white))

gr combine PA PB, rows(1) col(2) ycommon graphregion(color(white) fcolor(white)) ysize(7.5) xsize(11)


* *****

* FIGURE A2 -- PRED POS DISTN

* *****

use $all_ill_prob, clear

drop if mal_prob2 ==.

```

```

keep if first_ep

qui sum mal_prob2 if wt5_tertile_NoFever_rel1==1

    local lim1=r(max)

qui sum mal_prob2 if wt5_tertile_NoFever_rel2==1

    local lim2=r(max)

    kdensity mal_prob2, xline(`lim1', lcol(gs13)) xline(`lim2', lcol(gs13)) xtit("Predicted Malaria
Positivity", size(medsmall)) tit(" ") note(" ") graphregion(color(white) fcolor(white))

* *****

* FIGURE A3 -- PRICE HISTOGRAMS

* *****

use $pharma_txn, clear

histogram price_antimal if patient_age>=14, xline(40 60 100) xlab(50 "50" 100 "100" 150 "150" 200
"200" 250 "250+") xtit("Antimalarial Price (Ksh)") ///

    graphregion(color(white) fcolor(white)) tit("Ages 14+", size(medsmall)) name(p4, replace)

histogram price_antimal if patient_age>=9 & patient_age<=13, xline(30 45 75) xlab(50 "50" 100 "100"
150 "150" 200 "200" 250 "250+") xtit("Antimalarial Price (Ksh)") ///

    graphregion(color(white) fcolor(white)) tit("Ages 9-13", size(medsmall)) name(p3, replace)

histogram price_antimal if patient_age>=4 & patient_age<=8, xline(20 30 50) xlab(50 "50" 100 "100"
150 "150" 200 "200" 250 "250+") xtit("Antimalarial Price (Ksh)") ///

    graphregion(color(white) fcolor(white)) tit("Ages 4-8", size(medsmall)) name(p2, replace)

histogram price_antimal if patient_age<=3, xline(10 15 25) xlab(50 "50" 100 "100" 150 "150" 200 "200"
250 "250+") ///

    xtit("Antimalarial Price (Ksh)") graphregion(color(white) fcolor(white)) tit("Ages 3 and Under",
size(medsmall)) name(p1, replace)

gr combine p1 p2 p3 p4, col(2) row(2) graphregion(color(white) fcolor(white)) xsize(6) ysize(7)

```